

Introduction to Computing Hardware

1.1 What is Hardware?

- **Definition:** Physical components you can touch
- **Relationship:** Hardware needs software to function
- **Examples:** Monitor, keyboard, processor, memory

1.2 Types of Computing Devices

Desktop Computers:

- Stationary workstations
- Separate components (monitor, keyboard, mouse, system unit)
- More powerful processing capabilities
- Easier to upgrade and repair
- Better cooling systems

Laptops:

- Portable all-in-one design
- Built-in screen, keyboard, trackpad
- Battery-powered mobility
- Integrated components harder to upgrade
- Heat management challenges

Tablets:

- Touchscreen interface primary input method
- Lightweight and highly portable
- App-based ecosystem
- Limited multitasking compared to computers
- Good for media consumption and light productivity

Smartphones:

- Always-connected mobile devices
- Multiple functions: communication, computing, entertainment
- Small form factor with powerful capabilities
- Extensive sensor integration (GPS, camera, accelerometer)

1.3 Integrated and External Equipment

Input Devices:

- **Keyboard:** Text entry, shortcuts, navigation

- **Mouse/Trackpad:** Pointing, clicking, dragging
- **Touchscreen:** Direct manipulation interface
- **Camera:** Visual input for photos, video calls, scanning
- **Microphone:** Audio input for recording, voice commands
- **Scanner:** Convert physical documents to digital format

Output Devices:

- **Monitor/Screen:** Visual display of information
- **Speakers:** Audio output for media and notifications
- **Printer:** Create physical copies of digital documents
- **Projector:** Display to large screens or walls

Input/Output Devices:

- **Headset:** Audio input and output combined
- **External Storage:** Data input and output
- **Networking Equipment:** Data communication

Ports, Connections

2.1 Common Input/Output Ports

USB (Universal Serial Bus):

- Most common connection standard
- Multiple versions: USB 2.0, 3.0, 3.1, USB-C
- **USB 2.0:** 480 Mbps, black connectors
- **USB 3.0/3.1:** 5+ Gbps, blue connectors
- **USB-C:** Reversible, fastest speeds, power delivery
- Used for storage, input devices, charging, data transfer

HDMI (High-Definition Multimedia Interface):

- Digital audio and video transmission
- Supports high-definition and 4K resolution
- Common on TVs, monitors, projectors
- Single cable for both audio and video

VGA (Video Graphics Array):

- Older analog video standard
- 15-pin connector, usually blue
- Lower quality than digital connections
- Still found on older equipment

Other Common Ports:

- **Ethernet:** Wired network connection
- **Audio Jack:** 3.5mm for headphones/microphones
- **Power Jack:** Electrical power connection
- **DisplayPort:** Digital video alternative to HDMI

Practical Component (1 hour 15 minutes)

Practical Exercise 1.1: Hardware Identification Scavenger Hunt (30 minutes)

1. **Computer Tour:** Students examine their workstation
2. **Device Inventory:** Create list of all visible hardware components
3. **Port Investigation:** Locate and identify different ports
4. **Photo Documentation:** Take pictures of different hardware components
5. **Group Discussion:** Share findings and discuss component functions

Practical Exercise

1. **Hands-on Experience:** If available, interact with desktop, laptop, tablet, smartphone
2. **Comparison Chart:** Create detailed comparison of device types
3. **Use Case Analysis:** Determine best device for different scenarios
4. **Performance Testing:** Basic speed and capability comparisons

Assessment Questions:

- What are three main differences between desktops and laptops?
- Identify the input/output ports on your computer and explain their uses
- Which device would be best for a traveling salesperson and why?